

ATUL KUMAR

✉ Email | 🐙 Github | 🌐 Portfolio | 🔗 LinkedIn | </> LeetCode | </> Codechef

EDUCATION

Birla Institute of Technology, Mesra <i>Bachelor of Technology in Chemical Engineering</i>	2023 – Present Ranchi, India
Chinmaya Vidyalaya, Bokaro Steel City <i>Higher Secondary: Class: 11,12</i>	2021 – 2022 Bokaro, Jharkhand

EXPERIENCE

Web Development Intern <i>Zaalima Development (Remote)</i>	Apr 2026 – Jun 2026
- Developed an AI-powered Chrome Extension Generator using React, TypeScript, and Tailwind CSS, automating the generation of 5+ core extension files from natural language prompts and reducing project setup time by 80%. - Implemented a modular code generation engine supporting 20+ Chrome Extension APIs with reusable templates and validation, enabling rapid creation of production-ready browser extensions. - Owned the design and delivery of an enterprise SOP knowledge assistant (RAG-based), from retrieval architecture through deployment, cutting manual SOP lookup time for internal teams.	

PROJECTS

Stock-Research-Assistant <i>Python, FastAPI, LLMs, Financial APIs</i> 🐙	Jul 2026
- Developed a Stock Research Assistant integrating 4+ analysis modules for technical and fundamental analysis. - Built a FastAPI backend with 10+ REST API endpoints for automated stock data processing and AI-driven research reports. - Implemented LLM-based summarization analyzing 8+ financial indicators, reducing manual research time by 70%. - Designed a modular architecture integrating external APIs for real-time market data and explainable insights.	
SOP-RAG-Assistant <i>FastAPI, MongoDB, LLMs, Vector Embeddings</i> 🐙	May 2026
- Built an enterprise RAG assistant using FastAPI and LangChain with a 5-stage retrieval pipeline for SOP search. - Implemented semantic search via vector embeddings, improving response relevance by 40% over keyword baseline. - Built an end-to-end document pipeline (PDF parsing → chunking → embedding generation), reducing retrieval latency by 35%. - Enforced source-grounded response generation via retrieval-constrained prompting, ensuring every answer traces to a cited SOP section.	
Dental Caries Segmentation <i>YOLOv8, PyTorch, Computer Vision</i> 🐙	Jan 2026
- Developed a YOLOv8-based segmentation pipeline for detecting dental caries from 750+ radiographic images. - Trained on 768×768 images using AdamW optimizer with cosine learning-rate scheduling. - Applied augmentation techniques to increase effective dataset diversity by approximately 4×. - Achieved 22.6% mask recall using Precision, Recall, and mAP evaluation metrics.	

TECHNICAL SKILLS

LLM & GenAI: LangChain, Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, Prompt Engineering, LLM Integration
Machine Learning & Computer Vision: PyTorch, Scikit-learn, YOLOv8, Computer Vision, Object Detection, Image Segmentation, Natural Language Processing
Programming Languages: Python, C++, C, JavaScript, HTML, CSS
Backend & APIs: FastAPI, REST APIs
Databases: MongoDB, MySQL, PostgreSQL
Tools: Git, GitHub, Jupyter Notebook

CERTIFICATIONS & JOB SIMULATIONS

Deloitte Data Analytics Job Simulation (Forage)
Data Analyst Bootcamp (Udemy)